

Phase 4 Kick-off: Reviewing Phases 1–3 and Building the Capstone Bridge

OSALP 2025-04 Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

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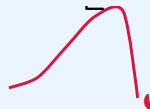
Gifted Education Fund: Off-school Advanced Learning Programmes
Learning Support Specialist: Mr Michael Kan

12 August 2026

The High-Stakes Hook: When Models Fail

❓ Opening Problem: The "Billion-Dollar Mistake"

In 1998, Long-Term Capital Management (LTCM), a hedge fund managed by Nobel laureates, collapsed and required a \$3.6 billion bailout. Their models were mathematically brilliant but failed to account for extreme real-world market panic and liquidity freezes.



🔑 Key Definition / Result: The Takeaway

This is what happens when you just calculate numbers without understanding the bridge to reality. Your Capstone Project is about avoiding this. You must connect your calculations to real-world risk management and honest assumptions.

Why This Review Matters Today

Opening Problem: From learning to project ownership

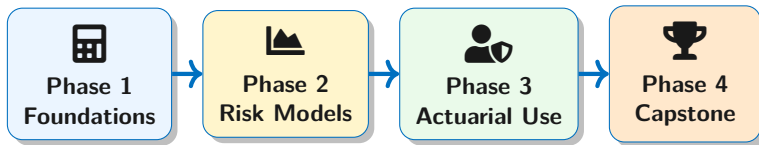
Students have completed a sequence of learning experiences in mathematics, statistics, finance, risk management, actuarial science, insurance, data tools, and responsible AI use. Phase 4 now asks one professional question: *Can your group turn learning into a defensible risk-management or actuarial project?*

Key Definition / Result: Today's purpose

This review is not a memory test. It is a bridge from Phases 1–3 into the Capstone Project. Students should leave with a clearer sense of:

- ▶ what tools they have already learned;
- ▶ what kinds of evidence their project should produce;
- ▶ how to connect data, simulation, assumptions, and interpretation;
- ▶ how to choose between fund risk analysis and insurance or annuity innovation.

The Big Learning Arc: From Uncertainty to Professional Judgement



Coherent pathway

- ▶ Phase 1 built the language of data, probability, statistics, finance basics, and discrete risk models.
- ▶ Phase 2 turned risk into a measurable modelling problem using portfolio diversification, distributions, Value at Risk, derivatives, hedging, and scenario analysis.
- ▶ Phase 3 connected probability, finance, and judgement to insurance, actuarial pricing, pooling, annuities, life contingencies, survival models, and actuarial work.
- ▶ Phase 4 requires students to combine these ideas into a project question, real data plan, simulation plan, responsible AI plan, and presentation message.

Phase 4 Is Also a Risk-Management Problem

A new kind of risk

Risk in Phase 4 is not only market risk, claim risk, mortality risk, or investment risk. It is also the uncertainty created by groupmates, communication, topic scope, data availability, simulation quality, AI verification, deadlines, and presentation readiness.

- ▶ **Risk event:** missing data, unclear project question, weak assumptions, unverified AI output, poor division of work, or slides without a clear story.
- ▶ **Consequence:** the final presentation becomes hard to defend even if students used many formulas.
- ▶ **Control action:** define a precise question, allocate roles, record sources, verify calculations, and rehearse the explanation.

Technology Check: Your Cheat Code

Think of this slide as your "cheat sheet" for not failing this project. Last year, groups lost points because they ignored team dynamics or didn't define their risks. This is your insurance policy!

Today's Output: What Students Should Decide

Investigation / Activity: Before leaving the kick-off

Each group should aim to decide the following initial items.

1. A group number and a reliable contact person.
2. A preferred project direction: fund risk analysis or insurance/annuity product design.
3. One possible project question, not only a vague title.
4. A first data plan: where real data may come from and what variables are needed.
5. A first simulation plan: what will be simulated and why the simulation helps.
6. A responsible AI plan: what AI may assist with and how students will verify it.

Technology Check: Bridge to Capstone

A strong Capstone Project starts with a good question, then builds evidence around that question. The project should not start from decoration, a generic title, or a copied AI answer.

Phase 1 Review: Foundations for Risk Thinking

Phase 1: What Students Were Introduced To

Foundational toolbox

Phase 1 built the mathematical and statistical foundations needed before actuarial or risk-management decisions can be discussed responsibly.

- ▶ Arithmetic, algebra, money, saving, and interest.
- ▶ Statistics foundations: data types, data collection, mean, median, mode, variability, surveys, and graphs.
- ▶ Probability basics: experiments, outcomes, sample spaces, events, rules, random variables, and scenarios.
- ▶ Discrete distributions and R coding with Google Gemini assistance.
- ▶ Finance introduction: markets, stocks, bonds, risk-return trade-off, and portfolio simulations.

Technology Check: Bridge to Capstone

For Capstone work, Phase 1 provides the language needed to define uncertainty clearly before calculating anything.

Phase 1 Skill 1: Turning Vague Uncertainty into Defined Events

Key Definition / Result: Core language

Before a calculation is meaningful, students must identify:

- ▶ the experiment or process with an uncertain outcome;
- ▶ the sample space of possible outcomes;
- ▶ the event that matters for the decision-maker;
- ▶ whether the event is defined by history, by a model, or by both.

Capstone translation

A vague statement such as *“this fund is risky”* should become a precise question such as *“How often did this fund lose more than 5% in a month, and how severe were the worst losses?”*

Common Mistake: Pitfall

Students should avoid treating ordinary-language phrases such as *safe*, *risky*, *stable*, or *good return* as if they were already mathematical definitions.

Phase 1 Skill 2: Probability Rules Are Modelling Choices

Rules students can reuse

- ▶ Complement: $P(A^c) = 1 - P(A)$.
- ▶ Addition law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- ▶ Conditional probability: $P(B | A) = \frac{P(A \cap B)}{P(A)}$, where $P(A) > 0$.
- ▶ Independence: $P(A \cap B) = P(A)P(B)$ only when one event does not change the probability of the other.

In Capstone work, these rules help students avoid double-counting, justify conditional reasoning, and explain why dependence or correlation may weaken simple models.

Phase 1 Skill 3: Counting Before Probability

Opening Problem: Why counting matters

In a discrete setting, probability is attached to a sample space. If the sample space is wrong, the probability calculation may look correct while describing the wrong world.

Key Definition / Result: Useful Capstone habit

Before applying a formula, ask:

1. What exactly is being counted or measured?
2. Does order matter?
3. Is the problem a path, a selection, a coded outcome, or a random variable?
4. Are outcomes equally likely, or do they need different probability weights?

Technology Check: Bridge to Capstone

For insurance or annuity innovation, counting possible claim patterns or benefit states helps students design the structure before estimating probabilities.

Phase 1 Skill 4: Random Variables and Distributions

From stories to numbers

A random variable turns a verbal uncertainty into a quantity that can be graphed, averaged, simulated, and compared.

- ▶ Claim frequency is often discrete.
- ▶ Claim severity may be continuous or represented by a discrete payout table for beginner modelling.
- ▶ A valid probability distribution needs non-negative probabilities that sum to one.
- ▶ Tables, bar charts, formulas, and interpretation sentences each reveal different modelling issues.

Capstone reuse

For a proposed phone-protection or travel-delay product, students can define X as the payout per customer, list possible values, assign probabilities from data or justified assumptions, and compute expected payout.

Phase 1 Skill 5: Expected Value, Variance, and Spread

 **Key Definition / Result:** Mean is not enough

$$E[X] = \sum_x xP(X = x),$$
$$\text{Var}(X) = \sum_x (x - \mu)^2 P(X = x), \quad \mu = E[X].$$

Expected value supports pricing and average cost thinking. Variance and standard deviation support risk, solvency, capital buffer, and short-run fluctuation thinking.

 **Common Mistake:** Capstone warning

A project should not conclude that one fund, strategy, or product is better only because it has a higher average return or lower average cost. Students should also compare variability, downside scenarios, and model assumptions.

Phase 1 Skill 6: Technology as a Modelling Assistant

Technology Workflow

1. Use spreadsheets or R to organise data, compute summaries, and draw charts.
2. Use Google Gemini to generate example code, suggest formulas, or explain a modelling idea.
3. Verify every result by checking assumptions, units, formulas, and whether the output makes sense.
4. Record prompts, datasets, code, assumptions, and verification decisions for project documentation.

Technology can make a wrong model look polished. In Phase 4, students must understand and verify their work, not only produce attractive outputs.

Phase 1 to Capstone: Practical Bridge

Phase 1 learning	Capstone use	Evidence students can show
Probability language	Define exactly what uncertain event matters.	Clear project question, event definitions, and assumptions.
Statistics summaries	Describe real data before modelling.	Descriptive table, chart, and interpretation sentence.
Random variables	Convert losses, returns, or benefits into measurable quantities.	Definition of X , support, probabilities, and units.
Expectation and variance	Compare average outcome and fluctuation.	Mean, standard deviation, downside explanation.
R, Excel, Sheets, Gemini	Analyse, simulate, visualise, and verify.	Code, formulas, charts, prompts, verification notes.

Interactive Challenge: Spot the Error

Common Mistake: A Dangerous AI Interpretation

 *Fictional AI Output:* "Because the historical average return of Fund A is 8% and Fund B is 6%, Fund A is strictly better and 100% of the portfolio should be allocated to it. There is no risk of losing money over the long term."

Investigation / Activity: Class Discussion: What is wrong here?

- ▶ Where is the variance or downside risk?
- ▶ What about correlation and diversification during a crisis?
- ▶ Does a historical average guarantee future performance?

Phase 2 Review: From Risk Concepts to Quantitative Modelling

Phase 2: What Students Were Involved In

Risk Management Science

Phase 2 developed students' ability to quantify financial and portfolio risk.

- ▶ Risk introduction and portfolio diversification.
- ▶ Probability distributions, including normal, binomial, and Poisson examples.
- ▶ Value at Risk and simulated loss/return distributions.
- ▶ Financial derivatives and risk hedging.
- ▶ R portfolio analysis, R simulations, Google Gemini code generation, result interpretation, and group discussions.
- ▶ Summative MCQ and industry-oriented learning through guest lecture materials.

Technology Check: Bridge to Capstone

For Capstone work, Phase 2 provides the quantitative tools to compare return, risk, downside exposure, and hedging logic.

Phase 2 Skill 1: Markets Create Prices, Liquidity, and Risk Transfer

Key Definition / Result: Market functions

A financial market is not only a place for trading. For risk managers and actuaries, markets support:

- ▶ **price discovery**, so portfolios and liabilities can be valued;
- ▶ **liquidity**, so assets can be bought or sold when cash is needed;
- ▶ **risk transfer**, so exposures can be shifted, shared, hedged, or rebalanced.

Capstone fund project

If a group analyses a real fund, they should ask not only whether the past return was high, but also whether holdings, market liquidity, and risk exposures support the fund's stated purpose.

Phase 2 Skill 2: Financial Instruments Have Different Risk Meanings

Instrument	Cash-flow idea	Risk-management interpretation
Debt / bonds	Contractual payments and principal repayment.	Useful for liability support, but still exposed to interest-rate, inflation, credit, and reinvestment risks.
Equity	Residual claim on business outcomes.	Higher uncertainty and potential upside; useful for growth but exposed to volatility and business risk.
Derivatives	Value linked to an underlying asset, rate, index, or price.	Can reshape exposure through hedging, exposure adjustment, or tactical risk transfer.



Common Mistake: Pitfall

Students should not automatically describe derivative use as speculation. A derivative can be speculative, but in risk management it is often used to reduce an existing exposure.

Phase 2 Skill 3: Time Value of Money Links Today and the Future

Key Definition / Result: Valuation engine

$$FV = PV(1 + i)^n,$$
$$PV = \frac{FV}{(1 + i)^n}.$$

Time value of money allows students to compare cash flows occurring at different dates. This is essential for bonds, reserves, annuities, insurance promises, and long-term project designs.

Capstone insurance product

If a proposed product pays future benefits, students should not compare future benefit amounts directly with today's premium without discounting or explaining the timing of cash flows.

Phase 2 Skill 4: Portfolio Risk and Diversification

Professional question

The professional question is not simply “*Which asset gives the highest return?*” A stronger question is “*What asset structure best supports the timing, liquidity, and risk characteristics of the liability or objective?*”

- ▶ Diversification can reduce risk when losses or returns are not perfectly linked.
- ▶ Correlation matters. If risks move together during stress, diversification may be weaker than the average-period chart suggests.
- ▶ A fund analysis project should examine both normal conditions and stressed downside situations.

Phase 2 Skill 5: Value at Risk and Scenario Thinking

Key Definition / Result: Risk measure

Value at Risk is a way to summarise downside risk at a chosen confidence level and horizon. It is a useful communication tool, but it depends on data, modelling assumptions, the selected horizon, and the shape of the loss distribution.

Modelling decision

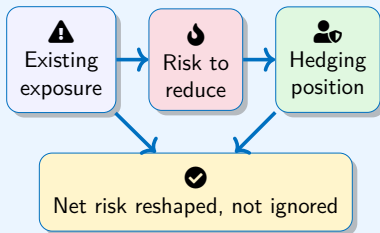
Choice. Students must decide the time interval, confidence level, data window, and return definition before calculating a risk measure.

Why this choice matters. Different choices can give different-looking risk numbers even for the same fund. A defensible project explains these choices instead of hiding them.

Phase 2 Skill 6: Hedging Is Deliberate Risk Transformation

Hedging logic

A hedge should be explained relative to an existing exposure.



Technology Check: Bridge to Capstone

In a fund risk project, students may discuss how diversification or hedging could reduce a specific exposure. In an insurance product project, students may discuss how deductibles, limits, exclusions, reinsurance-like logic, or product design reshapes risk.

Phase 2 to Capstone: Practical Bridge

Phase 2 learning	Capstone use	Evidence students can show
Portfolio diversification	Compare funds, asset classes, or simulated portfolios.	Return chart, risk chart, correlation discussion, scenario results.
Distributions and simulation	Model possible future returns or losses.	Distribution plot, simulated scenarios, assumptions.
Value at Risk	Summarise downside risk.	VaR table with horizon and confidence level.
Derivatives and hedging	Explain risk transformation.	Hedge rationale, before-and-after risk explanation.
Time value of money	Value future payments today.	PV/FV calculation, discount-rate assumption, timing diagram.

Phase 3 Review: Actuarial Science, Insurance, and Professional Context

Phase 3: What Students Were Involved In

Actuarial Science and Insurance Principles

Phase 3 connected mathematics, finance, risk sharing, and professional judgement.

- ▶ Actuarial overview and insurance basics.
- ▶ Expected loss, premium logic, pooling, and the Law of Large Numbers.
- ▶ Annuities, present value, life contingencies, and mortality tables.
- ▶ Survival models and insurance pricing.
- ▶ Excel risk/probability calculations, R curves and premium simulations, and Google Gemini assistance.
- ▶ Guest lecture exposure to actuarial roles, insurance sector work, qualification pathways, and interest-rate challenges.

Technology Check: Bridge to Capstone

For Capstone work, Phase 3 provides the insurance and actuarial interpretation needed to turn calculations into sustainable product or risk decisions.

Phase 3 Skill 1: From Uncertainty to Loss Exposure



Key Definition / Result: Insurance begins with financial consequence

Uncertainty alone is not enough for insurance thinking. Students must ask whether the uncertainty creates a possible financial loss that can be described, estimated, pooled, priced, or controlled.

Everyday phrase	Actuarial language	Capstone implication
"My screen might break." "How likely?"	Loss exposure. Probability or frequency.	Define the event and covered loss. Estimate from data or justified assumption.
"How expensive?" "Can we share it?"	Severity. Pooling or insurance.	Define benefit amount, cost, or payout. Judge whether many exposures can support the promise.

Phase 3 Skill 2: Risk Treatment Before Insurance

Four responses to risk

-  **Avoidance:** do not take the activity or exposure.
-  **Loss control:** reduce the chance or size of loss.
-  **Retention:** keep the risk and pay losses yourself.
-  **Transfer:** shift part of the financial burden to another party.

Insurance is a specialised risk transfer mechanism. A good product design may also include loss control, deductibles, limits, exclusions, waiting periods, or clear eligibility rules.

Phase 3 Skill 3: Pooling and the Law of Large Numbers

Key Definition / Result: Why pooling can work

A useful pool often needs many exposure units, losses that are not all triggered together, measurable costs, and a premium that people can realistically afford.

Common Mistake: Correlation warning

If a single event can damage many policyholders at the same time, pooling becomes weaker. The average may no longer be stable enough for ordinary insurance pricing.

Technology Check: Bridge to Capstone

In an insurance or annuity innovation project, students should explain why the proposed risk can or cannot be pooled. Climate, cyber, pandemic, and market-crash risks require special attention because dependencies can be strong.

Phase 3 Skill 4: Expected Loss, Premium, and Loading

Key Definition / Result: Pricing skeleton

$$\begin{aligned}\text{Expected Loss} &= \text{Frequency} \times \text{Severity}, \\ \text{Premium} &\approx \text{Expected Loss} + \text{expenses} \\ &\quad + \text{risk margin} + \text{profit or contingency loading}.\end{aligned}$$

Capstone product design

A proposed insurance product should justify the benefit design using expected cost, variability, affordability, and sustainability. A product that looks attractive to customers may still fail if pricing ignores severe or correlated losses.

Phase 3 Skill 5: Annuities, Mortality, and Time

Long-term promises

An annuity or life-contingent product links payment timing, survival or mortality assumptions, discounting, and product design.

- ▶ Present value compares future payments on a common date.
- ▶ Mortality tables and survival models translate life uncertainty into probabilities.
- ▶ Reserves support promises that continue after premiums are collected.
- ▶ Product design must make the promise understandable and financially credible.

Technology Check: Bridge to Capstone

If a Capstone group designs an annuity product, the project should clearly connect payment structure, survival assumptions, interest-rate assumptions, and sustainability.

Phase 3 Skill 6: Actuarial Roles and Professional Judgement

Actuaries use mathematics, statistics, financial theory, business judgement, and communication to study uncertain future events and help organisations plan and pay for them.

- ▶ Pricing actuaries help determine appropriate premiums.
- ▶ Reserving actuaries estimate future claims and liabilities.
- ▶ Financial modelling actuaries assess financial health and scenario impacts.
- ▶ Product development actuaries design competitive, sustainable, and regulatory-aware products.
- ▶ Risk management actuaries identify, assess, and manage risks.

Professional judgement means explaining assumptions, limitations, fairness, sustainability, and communication, not only showing a formula.

Phase 3 to Capstone: Practical Bridge

Phase 3 learning	Capstone use	Evidence students can show
Loss exposure	Define what can go wrong financially.	Covered event, excluded event, loss definition.
Pooling	Judge whether risk can be shared.	Exposure count, dependence discussion, pooling explanation.
Expected loss and premium	Build pricing logic.	Frequency, severity, expected cost, loading discussion.
Annuities and PV	Value long-term payments.	Cash-flow timeline, PV calculation, interest assumption.
Survival and mortality	Model life-contingent uncertainty.	Mortality or survival assumption and sensitivity check.
Professional roles	Communicate like a risk professional.	Clear assumptions, limitations, recommendation, and takeaway.

Bridge to Capstone: From Toolbox to Project Design

A Good Capstone Starts from a Good Question

Central Question

What risk-management or actuarial question can our group investigate with real data, simulation, clear assumptions, and a memorable conclusion?

Investigation / Activity: Question quality test

A good project question should be:

- ▶ clear enough to be answered within the programme schedule;
- ▶ interesting enough to motivate the audience;
- ▶ supported by real data and, where appropriate, simulation;
- ▶ linked to risk management or actuarial interpretation;
- ▶ honest about assumptions and limitations.

Two Approved Project Directions

Option 1: Fund risk analysis

Analyse one existing investment fund, or compare more funds if useful.

- ▶ Possible question: Is Fund A attractive after considering both return and risk?
- ▶ Natural tools: return calculation, volatility, drawdown, VaR, scenario simulation, charts, and interpretation.

Option 2: Insurance or annuity innovation

Invent an innovative insurance or annuity product supported by data and simulation.

- ▶ Possible question: Can we design a product that manages a real risk sustainably?
- ▶ Natural tools: loss exposure, frequency, severity, expected loss, premium logic, pooling, PV, and limitations.

3-Minute Challenge: The Worst Possible Idea

Icebreaker ❄️

Before we design a *good* project, let's understand the boundaries by designing a *terrible* one!

Investigation / Activity: Your Task (3 Minutes) ⌚

Talk to your group and invent the **worst possible** insurance product or fund analysis project.

 Example: "An insurance policy against alien abductions 🛸, priced at \$1, but we have no data and don't know what an alien is."

Why do this? It gets us laughing, talking, and paradoxically helps us realize exactly why real data, clear definitions, and feasible scope are mandatory!

Bridge Map for Option 1: Fund Risk Analysis

Project part	What students do	Which phase supports it
Question	Define the fund, benchmark, time period, and risk objective.	Phase 1: precise event/question language.
Data	Collect real price or return data and clean it.	Phase 1: statistics and data summaries.
Return	Compute simple or log returns and describe trend.	Phase 2: finance and valuation language.
Risk	Compute volatility, downside loss, drawdown, or VaR.	Phase 2: distributions, simulation, VaR.
Scenario	Simulate or stress test future returns.	Phase 2: scenario and simulation thinking.
Conclusion	Give a risk-adjusted recommendation with limitations.	Phase 3: professional judgement and communication.

Bridge Map for Option 2: Insurance or Annuity Innovation

Project part	What students do	Which phase supports it
Risk definition	Identify the loss exposure and covered event.	Phase 1 and Phase 3.
Data	Find real data for frequency, severity, mortality, or usage pattern.	Phase 1: data and summaries.
Model	Define random variables for claim count, claim amount, or benefit payment.	Phase 1: random variables and distributions.
Pricing	Estimate expected cost, premium logic, and sustainability.	Phase 3: expected loss and premium.
Timing	Discount future payments if benefits occur later.	Phase 2 and Phase 3: TVM and PV.
Reality check	Discuss pooling, adverse selection, moral hazard, correlation, and limitations.	Phase 3: insurance judgement.

Minimum Evidence Standard for Any Capstone

Key Definition / Result: Evidence before persuasion

A presentation becomes convincing when it shows evidence, not just an idea.

1. **Real data:** tell the audience what data you used and why it is relevant.
2. **Clear assumptions:** state assumptions in ordinary language and mathematical form where possible.
3. **Simulation:** explain what was simulated, why it matters, and what the result means.
4. **Charts or tables:** show patterns, not only final numbers.
5. **Interpretation:** connect results back to risk management or actuarial judgement.
6. **Limitations:** say what your model cannot prove or what could change the conclusion.

Responsible AI Use in Phase 4

AI Exploration: Allowed support, with verification

Students may use Google Gemini to explore public datasets, generate ideas, draft code, support Google Apps Script, help create HTML Canvas apps, summarise, polish, or visualise.

Common Mistake: Non-negotiable rule

AI can assist, but students must understand, verify, and cite their work. Avoid direct copy-and-paste.

Technology Workflow

- ▶ Keep a record of prompts and AI-generated code or text.
- ▶ Run and inspect calculations rather than trusting outputs silently.
- ▶ Compare code results with a small manual calculation when possible.
- ▶ Cite data sources and distinguish student work from AI assistance.

Technology Check: The AI Survival Guide

Instead of just "rules", view this as your cheat code for using AI safely. AI can hallucinate formulas. If you copy-paste blindly, you inherit its mistakes. Verify everything to protect your project's credibility.

What Makes the Presentation Effective

Keith-style phrase	Capstone meaning
To motivate me	Explain why the topic matters and what real risk is involved.
To surprise me	Show an interesting insight, comparison, invention, or unexpected finding.
To impress me	Support the message with data, simulation, evidence, and reasoning.
To present in multiple ways	Use charts, tables, formulas, examples, analogies, poster, video, or demo when useful.
To summarize	End with a clear takeaway the audience can remember.

Good presentation turns analysis into a story: question, evidence, insight, implication, and takeaway. 📖

Suggested Presentation Storyline

- ▶ **Motivation:** What real risk or decision does your project address?
- ▶ **Question:** What exactly are you investigating?
- ▶ **Data:** What data did you use, and what are its limitations?
- ▶ **Model:** What random variables, financial formulas, or actuarial logic did you use?
- ▶ **Simulation:** What future or uncertain outcomes did you simulate?
- ▶ **Results:** What did the numbers, charts, and scenarios show?
- ▶ **Interpretation:** What does this mean for risk management or actuarial decision-making?
- ▶ **Limitations and conclusion:** What should the audience remember, and what should not be overclaimed?

From Formula to Interpretation Sentence

Formula is not the final answer

A formula or chart is only part of the project. Students should add an interpretation sentence such as:

- ▶ *The simulated loss distribution suggests that average cost alone is not sufficient because the worst 5% of scenarios are much more severe.*
- ▶ *The fund had higher average return, but its downside risk was also higher during stress periods.*
- ▶ *The proposed premium is sustainable only if the claim frequency remains near the estimated range.*

If the audience cannot understand what a number means for a decision, the analysis is not yet complete. 💡

Group Roles: Useful, But Not Walls

Suggested role	Possible responsibility
Group coordinator	Keeps deadlines, communication, file organisation, and records.
Data lead	Finds data sources, checks data quality, and manages analysis files.
Actuarial/risk analysis lead	Connects the project to risk management or actuarial thinking.
AI/tools lead	Explores R, Excel, Google Sheets, Google Gemini, Apps Script, HTML Canvas, Python, or Colab.
Presentation lead	Plans storyline, speaking arrangement, and rehearsal.
Slides writer	Organises slides, visuals, charts, captions, and final formatting.

Roles are useful for organisation, but every member should understand the project question, data, simulation, findings, and presentation message.

Investigation / Activity: Immediate Action (2 Minutes)

Look at your group members right now. Take 2 minutes to tentatively claim a role. You can always change it later, but make a decision today so work can begin!

Proposal Submission: What Must Be Ready

Investigation / Activity: Proposal items

Each group proposal should include:

- ▶ group number and contact person;
- ▶ group members;
- ▶ project direction: fund risk analysis or insurance/annuity product;
- ▶ topic title;
- ▶ motivation;
- ▶ real data plan;
- ▶ simulation plan;
- ▶ responsible AI use plan;
- ▶ division of work;
- ▶ timeline.

Key Definition / Result: Deadline

Submit the written proposal to Mr Michael Kan, Learning Support Specialist, no later than **24 August 2026, Monday, 23:59**.

Phase 4 Schedule and Working Rhythm

Session	Date	Time	Mode	Main purpose
15	12 Aug 2026	13:30–16:00	CUHK	Kick-off, review Phases 1–3, group formation, topic selection.
16	26 Aug 2026	13:30–14:45	Zoom	Weekly progress check and learning support feedback.
17	9 Sep 2026	16:30–17:45	Zoom	Weekly progress check and learning support feedback.
18	23 Sep 2026	16:30–17:45	Zoom	Weekly progress check and learning support feedback.
19	7 Oct 2026	16:30–17:45	Zoom	Dry-run presentations and final adjustments.

Before You Leave Today

Investigation / Activity: Student checklist

- ▶ I know my group number.
- ▶ I know my groupmates.
- ▶ We have one possible topic.
- ▶ We know what data or examples we may need.
- ▶ We have assigned immediate next steps.
- ▶ We know what to do before the next session.

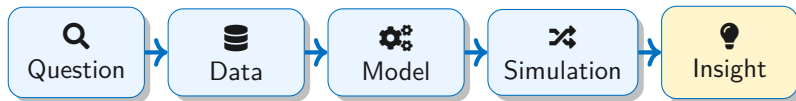
Technology Check: Bridge to Capstone

Phase 4 is where students move from learning examples to owning a project. A strong group acts like a small risk team: define the problem, gather evidence, model carefully, communicate clearly, and manage its own project risk.

Final Message: What You Have Learned Is Enough to Start

Key Definition / Result: Confidence with responsibility

Students do not need a perfect model on the first day of Phase 4. They need a serious question, a workable data plan, a simulation idea, and a willingness to verify and improve their assumptions.



A strong Capstone Project is not about using every tool. It is about choosing the right tools, explaining why they fit, and defending the decision honestly.

Thank You 😊

Phase 4 Capstone Project Development

OSALP 2025-04

Nurturing Future Risk Managers and Actuaries
Department of Statistics and Data Science, CUHK